International University of Business Agriculture and Technology

Assignment

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Submitted To:
Krishna Das
Associate Professor
Deptartment of Computer Science and Engineering
IUBAT

Submitted By:
Lamia Jamal
ID: 23103324
Section: F

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1. Comparison of indoor air quality and thermal comfort standards and variations in exceedance for school buildings

Title Description: Indoor air quality (IAQ) and thermal comfort are important in schools, but differences among standards can lead to inconsistent assessments. This study compares EN 16798-1, BB101, and ASHRAE 55 and 62.1 using environmental data from four schools in northern Italy. Air temperature, globe temperature, relative humidity, and CO_2 were measured to assess thermal comfort and IAQ. The study also examined how different building-material emission levels affect indoor pollutant concentrations. Results showed considerable differences among the standards in their assessment methods and threshold values. The study further identified the need for a combined IAQ and thermal comfort assessment approach to support better decisions on indoor environmental conditions and energy use.

Method Applied:

- Field Data Collection: Indoor environmental data were collected from four naturally ventilated schools in Bolzano, northern Italy, from February to April 2022. Two classrooms were monitored in each school using an in-house multi-sensor device called the Environmental Quality bOX (EQ-OX).
- Environmental Parameter Measurement: The EQ-OX system measured air temperature, globe temperature, relative humidity, air pressure, and CO_2 concentration. A gateway and 4G modem were used to transmit and store the measurements in real time.
- Sensor Validation and Data Processing: Before the monitoring campaign, the sensors were tested against calibrated reference instruments to verify their accuracy. The collected readings were resampled into 10-minute average values for analysis. Data processing and calculations were performed using Python, mainly with Pandas and Matplotlib.
- Standards-Based Assessment: Indoor environmental conditions were evaluated using EN 16798-1, BB101, ASHRAE 55, and ASHRAE 62.1 to assess IAQ and thermal comfort according to different regulatory criteria.

- Thermal Comfort Analysis: Thermal comfort was assessed using operative temperature and PMV. Mean radiant temperature was calculated from the measured air and globe temperatures, and operative temperature was then derived from air and mean radiant temperatures. PMV was calculated using assumed values for air speed, clothing insulation, and metabolic rate because air speed was not directly measured.
- IAQ Analysis: CO_2 concentration was used as the main indicator of IAQ and compared with the threshold values specified by the selected standards. The percentage of occupied time falling within different IAQ and thermal comfort categories was calculated.
- Emission Scenario and Contaminant Simulation: Three building-material emission scenarios, high-emitting, low-emitting, and very-low-emitting, were simulated to examine their potential effects on indoor formaldehyde and CO_2 concentrations. The CO_2 decay method was first used to estimate the air change rate of each classroom. CO_2 and formaldehyde concentrations were then estimated using a single-zone, fully mixed contaminant mass-balance model incorporating classroom volume, ventilation rate, occupancy, outdoor concentration, and pollutant emission rates. Formaldehyde emissions from building materials and furniture were assigned according to their emission classifications, and the total emission rate for each classroom was calculated based on the emission rate and surface area of the materials.

Gaps Identified/Scope:

- Existing IAQ and thermal comfort standards use different thresholds and assessment methods, which can produce inconsistent results.
- The study notes that current standards do not explicitly provide guidance on using building simulation methods for long-term or spatially detailed environmental assessment.
- The study was based on only four schools in northern Italy, so its findings cannot be directly generalized to all school buildings or climates.
- Existing standards generally assess IAQ and thermal comfort independently rather than as interconnected environmental conditions.

- The study focused on evaluating standards rather than developing a predictive Machine Learning (ML) model.

Critical Analysis: The study provides a useful assessment of how the choice of standard can affect the interpretation of indoor environmental quality. Its main strength is the use of measurements from occupied classrooms, allowing the assessment to reflect actual building conditions rather than relying entirely on theoretical assumptions. Considering both thermal comfort and IAQ also provides a more complete assessment of the indoor environment. However, the assessment is affected by uncertainties in the measured and assumed variables. For example, air speed was not directly measured and some comfort-related parameters were assumed, which can influence PMV calculations. The relatively short monitoring period and small number of schools also limit the ability to represent seasonal and climatic variations. The pollutant analysis introduces another limitation because formaldehyde concentrations were estimated through simulation rather than directly measured, meaning that the results depend partly on assumptions about emission rates and ventilation. These limitations indicate that standards-based assessment alone may not be sufficient for understanding how environmental conditions vary within different buildings and operating conditions. The study therefore highlights the value of combining measured environmental data, building characteristics, and predictive approaches to provide more detailed and adaptable assessments of indoor environmental conditions.

Source:

https://www.sciencedirect.com/science/article/pii/S2352710223005843/pdfft?md5=6d8040adff6ed1bed29f671f9defa1e7&pid=1-s2.0-S2352710223005843-main.pdf

2. Physics-based machine learning for predicting urban air pollution using decadal time series data

Title Description: The study investigates whether Physics-Based Machine Learning (PBML) can improve air-pollution prediction compared with LSTM and linear regression. It uses ten years of daily traffic, weather, and air-pollution data from three Norwegian cities to predict NOx and $PM_{2.5}$ concentrations.

Method Applied:

- Dataset and Study Area**:** The study uses daily time-series data from 2009 to 2018 for three Norwegian cities: Oslo, Bergen, and Trondheim. The dataset combines air-pollution, traffic, and meteorological observations. Air-quality data include $PM_{2.5}$ and NOx concentrations obtained from the Norwegian Institute for Air Research. Traffic volume data were obtained from the Norwegian Public Road Administration, while weather data were obtained from the Norwegian Meteorological Institute.
- Predictor Variables: Traffic and meteorological variables were used to explain variations in pollutant concentrations. The weather variables include mean air temperature, heating degree days (HDD), vapour pressure, mean wind speed, mean wind gust, relative humidity, snow depth, and precipitation**.** Previous pollutant concentrations were also considered when making future predictions.
- Data Splitting and Validation: Because the dataset is time-dependent, the observations were divided chronologically rather than randomly. The first 60% was used for training, while the remaining 40% was divided equally into validation and test sets, resulting in 60% training, 20% validation, and 20% testing. This approach prevents future observations from being used to train the model.
- Long Short-Term Memory (LSTM) & Linear Regression Model**:** An LSTM neural network was used to capture temporal dependencies in air-pollution data using historical observations from 1 to 7 previous days, with hyperparameters such as the number of LSTM units, dropout rate, and learning rate tuned on the validation set. Linear Regression was also used as a conventional statistical baseline to assess whether the more advanced LSTM and PBML approaches provided improved prediction performance.
- Physics-Based Machine Learning (PBML): The study developed a PBML model by combining a neural network with ordinary differential equations (ODEs) representing the physical processes affecting pollutant concentrations. The ODE framework was used to represent the temporal changes in NOx and $PM_{2.5}$ based on traffic and meteorological conditions. The neural network learns from the observed data while the physical component constrains the model according to the underlying pollutant dynamics.

- <u>Physical Constraints and Loss Function</u>**:** The PBML model combines the difference between predicted and observed pollutant concentrations with an ODE-based physics loss**.** This allows the model to consider both data-driven patterns and known physical relationships rather than relying only on observed correlations.
- <u>Model Evaluation</u>**:** The models were evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2. RMSE was used as the main prediction-error measure, while MAE assessed the average magnitude of prediction errors and R^2 measured how much variation in pollutant concentrations was explained by the model. Statistical tests were also used to examine differences between model performances.

Gaps Identified/Scope:

- The study mainly captures the temporal dimension because detailed spatial information was unavailable.
- The authors therefore used ODEs rather than PDEs and identified the need for datasets containing both spatial and temporal information.
- The study also focuses on NOx and $PM_{2.5}$ rather than CO_2.

Critical Analysis: The study makes a strong methodological contribution by incorporating physical relationships into the ML prediction process rather than relying only on statistical patterns in historical data. The use of chronological data splitting is also appropriate for time-series prediction because it reduces the risk of information from future observations influencing model training. The PBML approach provides a way to constrain predictions according to known pollutant dynamics, which can improve the physical relevance of the results. However, the physical component is represented through ODEs and therefore mainly describes temporal behavior. Urban pollution is also strongly affected by spatial transport, local emission sources, and differences between locations, which cannot be fully represented without spatial information. The use of daily observations may further reduce the model's ability to capture short-term changes in pollution caused by rapidly changing weather or traffic conditions. In addition, improvements in prediction error do not automatically establish that the physics-based approach is consistently superior, particularly where statistical tests show limited significant differences between models. The study therefore demonstrates the value of incorporating physical knowledge into ML while also showing that prediction quality depends on how completely the physical and environmental processes are represented. A more comprehensive prediction framework would benefit from higher-resolution spatial and temporal information while retaining physically meaningful constraints.

Source:

https://iopscience.iop.org/article/10.1088/2515-7620/add795/pdf

3. Enhancing urban air quality prediction using time-based-spatial forecasting framework

Title Description: The study addresses the challenge of accurately forecasting urban air quality by considering both spatial and temporal variations. The authors propose a Time-Based-Spatial (TBS) forecasting framework that combines machine learning with time-series modeling to predict the AQI. The study uses air-quality data from 22 cities in India, covering different geographic and environmental conditions. The framework aims to improve short-term AQI forecasting by considering differences between locations as well as changes in pollutant concentrations over time.

Method Applied:

- Hybrid CNN model: A Convolutional Neural Network (CNN) is used to process spatial information. Normalized latitude and longitude are incorporated to represent geographic differences between cities. A separate pollutant branch processes concentrations of $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3 and NH_3.
- ARIMA: An Autoregressive Integrated Moving Average model is applied to identify temporal patterns and trends in AQI. Individual ARIMA models are developed for each city to predict short-term variations in air quality.
- Spatial-temporal integration: The outputs of the CNN and ARIMA models are combined through weighted averaging to produce the final AQI forecast. This allows the framework to consider both geographic and time-dependent characteristics of air quality.

- Dataset: Air-quality data from 22 Indian cities were obtained from the Central Pollution Control Board (CPCB). The dataset contains timestamps, geographic coordinates, pollutant concentrations and AQI values.
- Forecasting: The proposed framework produces a 6-hour AQI forecast, targeting short-term air-quality monitoring and decision-making.

Gaps Identified/Scope:

- No prediction of individual pollutants such as CO_2.
- Spatial information is represented mainly through latitude and longitude, without detailed urban morphology, building geometry or other physical characteristics.
- Meteorological variables and other external factors, such as temperature, wind, humidity, industrial activity and traffic, were not fully incorporated.

Critical Analysis: The study makes an important methodological contribution by recognizing that urban air quality is influenced by both changes over time and differences between locations. The use of CNN to represent spatial information and ARIMA to capture temporal patterns provides a reasonable structure for combining these two dimensions. The framework is particularly useful for short-term forecasting because it can account for recent pollutant behavior while incorporating geographic information. However, representing spatial conditions mainly through latitude and longitude provides only a limited description of the urban environment. Geographic coordinates indicate where a measurement is located but do not explain important local factors such as building density, road configuration, traffic conditions, vegetation, or surrounding structures. The framework also does not fully incorporate meteorological variables that can influence pollutant dispersion and accumulation. In addition, predicting AQI rather than individual environmental parameters limits the ability to understand the specific behavior of each pollutant and other environmental conditions. These limitations suggest that spatial-temporal prediction becomes more informative when location is represented through meaningful physical and environmental characteristics rather than coordinates alone. The study therefore supports the need for more detailed spatial representations and additional environmental variables when developing localized prediction models.

Source:

https://www.nature.com/articles/s41598-024-83248-z#Abs1

4. Thermal and visual comforts of occupants for a naturally ventilated educational building in low-income economies: A machine learning approach

Title Description: This study develops machine learning models to predict thermal and visual comfort in a naturally ventilated educational building in Dhaka, Bangladesh. The study uses 1,310 cleaned field-survey samples containing demographic, building-related, environmental, and comfort variables. Environmental measurements include air temperature, humidity, and CO_2 concentration. The main focus is on comparing Random Forest (RF), Decision Tree (DT), and XGBoost models and examining how feature selection and imbalanced-data handling affect prediction performance.

Method Applied:

- Dataset and input features: Field data were collected from students in a naturally ventilated educational building in Dhaka. The dataset includes variables such as gender, age, room orientation, number of fans, lights, windows, floor area, distance from windows, temperature, humidity, and CO_2 concentration. Thermal comfort and visual comfort were used as prediction targets.
- Handling imbalanced data: SMOTE generated synthetic samples for minority classes, while Tomek Links and ENN removed problematic or noisy samples.
- Random Forest (RF): RF was used as an ensemble classification model. It constructs multiple decision trees and combines their predictions. The model was selected because it can capture nonlinear relationships and interactions between multiple environmental and building variables.
- Decision Tree (DT): DT was used as an interpretable classification model. It creates decision rules by recursively splitting the data based on input features. It was included as a simpler baseline for comparison with the ensemble models.

- XGBoost: XGBoost was applied as a gradient-boosted tree model. It builds trees sequentially, with each new tree attempting to improve the errors of previous trees.
- Feature selection: Recursive Feature Elimination with Cross-Validation (RFECV) was applied to determine which variables were most useful for each ML model. This is particularly relevant because the models did not necessarily require the same number of features.
- Model evaluation: The models were evaluated using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC.

Gaps Identified/Scope:

- The data were collected from one educational building in Dhaka, limiting generalizability.
- The study used only three ML algorithms, so other ML and deep learning models could potentially provide different or better results.
- The data were collected during a specific period and time of day, so the models may not fully capture seasonal and daily variations in environmental conditions.

Critical Analysis: The study demonstrates that ML can capture complex relationships between building characteristics, environmental conditions, and occupant comfort in naturally ventilated educational buildings. Its use of real field-survey data is a major strength because comfort responses are influenced by both measurable environmental conditions and occupant-related factors. The application of feature selection also provides useful evidence that the importance of input variables can differ according to the prediction task and model. However, the improvement obtained through data-balancing techniques shows that prediction performance is influenced not only by the choice of ML algorithm but also by the quality and distribution of the training data. This is important because high classification performance can otherwise give an incomplete impression of model reliability when some comfort categories are underrepresented. The study is also limited to one educational building and a specific data-collection period, which restricts its ability to represent different buildings, seasons, and environmental conditions. Furthermore, CO_2 is used as an input variable rather than being predicted as an environmental outcome, so the study does not directly address how environmental parameters themselves may change under different conditions. Overall, the study demonstrates the usefulness of ML for building-environment prediction but also indicates that reliable environmental modeling requires careful treatment of data quality, feature selection, and variation across time and building conditions.

Source:

https://www.sciencedirect.com/science/article/abs/pii/S2352710224015833

5. **Air quality prediction and control systems using machine learning and adaptive neuro-fuzzy inference system**

Title Description: The study investigates ML-based prediction of air pollutants in Semnan, Iran, where industrial development has contributed to increasing air pollution. The researchers evaluated several ML models for predicting $PM_{2.5}$ concentrations using pollutant monitoring data collected across different seasons. The study aimed to identify a reliable prediction model that could support air-quality monitoring and real-time decision-making.

Method Applied:

- Dataset: Air-quality monitoring data collected in Semnan from 2020 to 2021, containing NO_x, CO, NO, O_3, NO_2, SO_2 and $PM_{2.5}$ measurements.
- ML models: Random Forest (RF), Random Tree, M5P, Decision Stump, Multilayer Perceptron (MLP), Gaussian Process, Linear Regression, Simple Linear Regression and SMOreg.
- ANFIS: An Adaptive Neuro-Fuzzy Inference System was developed using a Sugeno-type fuzzy inference system and hybrid learning.
- Seasonal analysis: Models were evaluated separately across six seasonal datasets to examine changes in prediction performance.
- Evaluation metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Relative Mean Absolute Error (RMAE), Root Relative Mean Squared Error (RRMSE), and correlation coefficient of determination (R^2).

Gaps Identified/Scope:

- The study focuses mainly on $PM_{2.5}$ prediction and uses other pollutant concentrations as major input variables.
- The dataset is also relatively small compared with large-scale high-resolution environmental datasets.
- The study is based on a single city, so the generalizability of the models to other urban environments is uncertain.
- Model performance can depend strongly on the selected input variables and data-splitting strategy, so the extremely high R^2 values reported for ANFIS should be interpreted carefully.

Critical Analysis: The study demonstrates the ability of ML and ANFIS to model nonlinear relationships in air-quality data and provides useful evidence that prediction performance can vary across seasons. The seasonal evaluation is a strength because pollutant behavior is not constant throughout the year and a model that performs well under one set of environmental conditions may not maintain the same performance under another. However, the prediction framework relies heavily on other pollutant concentrations as input variables. While these variables can improve prediction accuracy, this approach provides less information about how meteorological and surrounding environmental conditions influence pollutant behavior. The relatively small dataset and single-city setting also limit the generalizability of the results. In addition, the very high R^2 values reported for ANFIS require careful interpretation because high predictive performance may partly reflect the selected input variables and their relationship with the target pollutant. Without broader environmental and spatial information, it is difficult to determine whether the model would maintain similar performance under substantially different conditions. The study therefore demonstrates the predictive capability of ML-based air-quality models while highlighting the importance of broader input variables, stronger validation, and more diverse datasets for developing models that can generalize beyond a specific location and dataset.

Source:

https://www.sciencedirect.com/science/article/pii/S2405844024158148/pdfft?md5=dd50773cde17d9ddede8b31fb6f5687a&pid=1-s2.0-S2405844024158148-main.pdf